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REVIEW ARTICLE

The effects of different pretreatment technologies on microbial community in anaerobic digestion process: A systematic review

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Abstract

Here we comprehensively review the available knowledge on effects of different pretreatment technologies on microbial population and microbial dynamics in anaerobic digestion (AD) fed with different substrates and different operational parameters. To identify peer-reviewed studies published in English-language journals, a comprehensive search was performed across multiple electronic databases. The eligible studies were analyzed to extract data and information pertaining to the configuration of anaerobic reactors, operational parameters, and various pretreatment processes such as chemical, biological, enzymatic, thermal, microaerobic, and ultrasonic. The findings derived from this current review demonstrated that different chemical, biological, and physical pretreatment technologies improve the biomethane potential (BMP) and potentially affect the dominant bacteria and archaea. Moreover, although hydrogenotrophic methanogenesis are more observed due to resistance to extreme conditions, methane production follows both aceticlastic and hydrogenotrophic pathways in AD assisted with different pretreatment process. *Firmicutes* and *Bacteroidetes* phyla of bacteria were the dominant hydrolytic bacteria due to synergetic effects of different pretreatment process on solubilization and bioavailability of recalcitrant substrates. In summary, a holistic understanding on bacteria and archaea communities, along with the mechanisms of the dominant microorganisms leads to enhanced stability and overall performance of anaerobic digestion (AD) processes.

Keywords Anaerobic digestion · Molecular techniques · Biomethane potential · Microbial population · Pretreatment process

Introduction

Global environmental issues and energy crisis are two major driving forces for researchers to vigorously develop the renewable and eco-friendly energy sources such as biomass-based energy which can also solve the carbon neutrality [1, 2]. The biomass-based energy and biogas production via anaerobic digestion (AD) process even overcome global warming and environmental pollution attributed to fossil fuels [3]. Over the recent decades, AD as the sustainable, economical, practical and proven biological technology has been extensively used to harness different aims including energy recovery from different organic substrates, sludge reduction and stabilization and, pathogen removal [4, 5]. These advantages are the main driving factors for the widespread adoption of AD [6–9]. AD, as complex biological process containing a mixed culture of microorganisms, convert the biodegradable waste into the biogas as the renewable energy source [10]. AD process consists of four consecutive phases: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. Each phase necessitates specific and appropriate bacteria and archaea to carry out the required transformations [4, 11]. The hydrolysis phase, as the initial step in the fermentation process plays important role in overall rate of the process; native microorganisms in conventional AD can't easily biotranformate and degrade the recalcitrant organic

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matters and complex substrates including lignocellulosic biomass into simple monomeric molecules which further converted to precursors of methanogenisis [4, 12]. Over the recent years, various physical, chemical, and biological methods have been utilized to surmount this obstacle, aiming to enhance the performance of AD. This, in turn, leads to improved biogas production and optimization of energy resources [1, 13]. The pretreatment process can facilitate the break down and bioavailabity of substrates for anaerobic microbes involved in AD process [14, 15]. Although some pretreatment technologies are questioned due to low technical feasibility, toxic by-products, high operational cost and waste discharged into environment, the higher BMP and energy balance and even pathogen removal can justify the wide application of these pretreatment process for improvement the AD [1, 16]. According to open literature review, many chemical [5], biological [17], enzymatic [10], thermal [18], ultrasonic [19] and microaerobic [20] pretreatment have recently been employed to enhance the AD performance and consequently BMP from different recalcitrant substrates. The interdependent relationship between microorganisms is generally influenced by operational and environmental parameters; the microorganisms tolerant to new and unfavorable conditions get predominant of microbial community involved in AD and handle the biogas production [21, 22]. Therefore, knowledge on the responses of microbial community and, specially, the dominant bacteria and archaea can potentially improve the process efficiency and stability [18, 19]. In addition, the emergence of new and innovative molecular technologies including next-generation sequencing (NGS) technologies, fluorescence in situ hybridization (FISH), and quantitative real-time polymerase chain reaction (qPCR) analysis aid the researchers to track the microbial dynamics and predominant microbial population in AD. After the advent of innovative molecular technologies, many studies have focused on the microbial community in AD assisted with different pretreatment technologies [9, 22, 23]. However, there are some sporadic and contradictory information on microbial population and dominant microorganisms in AD assisted with pretreatment technologies. Therefore, a comprehensive understanding of the composition and dynamics of the microbial community in AD with the assistance of various physical, chemical, and biological technologies can improve the understanding on stability and improvement of biogas production. Therefore, the present systematic review was designed for the first time to comprehensively survey roles of different chemical, biological, enzymatic, thermal, ultrasonic and microaerobic pretreatment in improvement of BMP in AD and investigate the corresponding microbial dynamics and predominant bacteria and arachaea involved in AD.

Approach

Literature search and inclusion criteria

The present study was performed to address the main question: Do pretreatment technologies influence the dominant bacteria and archaea in AD process fed with different substrate and different operational parameters?. In order to achieve this objective, the researchers utilized the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to guide the study design, search strategy, and study selection process [24]. Moreover, a comprehensive search was conducted across four prominent global electronic databases (Scopus, Web of Science, PubMed, and Embase) following their respective search guidelines to ensure the inclusion of relevant studies. Furthermore, for the rigorous selection process of reliable studies, Boolean operators 'AND' and 'OR' were used in conjunction with the following search terms: "Microbial community" OR "microbial population" OR "microbial diversity" OR "microbial dynamic*" AND "anaerobic digestion" OR "anaerobic reactor*". The comprehensive search strategy for different pretreatment process and their effects on microbial dynamics and microbial population are summarized in **additional file**. Moreover, the handsearching was used to identify the additional sources. The detailed information on the study selection process and reliability are reported in our previous study [9]. We included studies that were published in peer-reviewed journals and written in the English language, without any restrictions on the publication date. The selected studies encompassed microbial variations observed in full-scale, lab-scale, and experimental anaerobic bioreactors, regardless of the bioreactor configurations. However, conference papers, book chapters, and abstracts were excluded from our analysis.

Study selection and data extraction process

Once the duplicate studies from different electronic databases were eliminated and the research met the objectives and inclusion criteria, a thorough extraction of data from the eligible studies was conducted. The authors assessed both the general and specific information obtained from the included studies. This information contained the name of the first author, publication year, journal name, substrate characteristics, inoculum properties, and various operational parameters (such as operational temperature, anaerobic digestion mode, the pretreatment method and corresponding operational characteristics, Inoculum, Substrate and the highest BMP pretreatment). Moreover, the pertinent data concerning microbial population and dynamics, with particular emphasis on dominant bacteria and archaea, were extracted and systematically organized in appropriate tables.

Results and discussion

Literature search and study characteristics

Figure 1 displays the PRISMA flow chart, illustrating the quantity of articles identified from different electronic databases, the screening process, and the evaluation based on eligibility criteria and review objectives. Generally, 958 articles were identified from the different electronic databases, which reduced to 155 after the removal of duplicated studies. Upon reviewing the titles and abstracts of the identified articles, a total of 74 peer-reviewed articles were selected. Subsequently, after a thorough examination of the full articles, an additional 41 articles were excluded from the review due not elaborating the effects of pretreatment on microbial dominant and corresponding mechanism in AD. Finally, 32 articles focusing on the effects of different types of pretreatment process including chemical, biological, enzymatic, microaerobic, thermal, and ultrasonic on microbial composition, dominant bacteria and archaea in anaerobic digestion were selected.

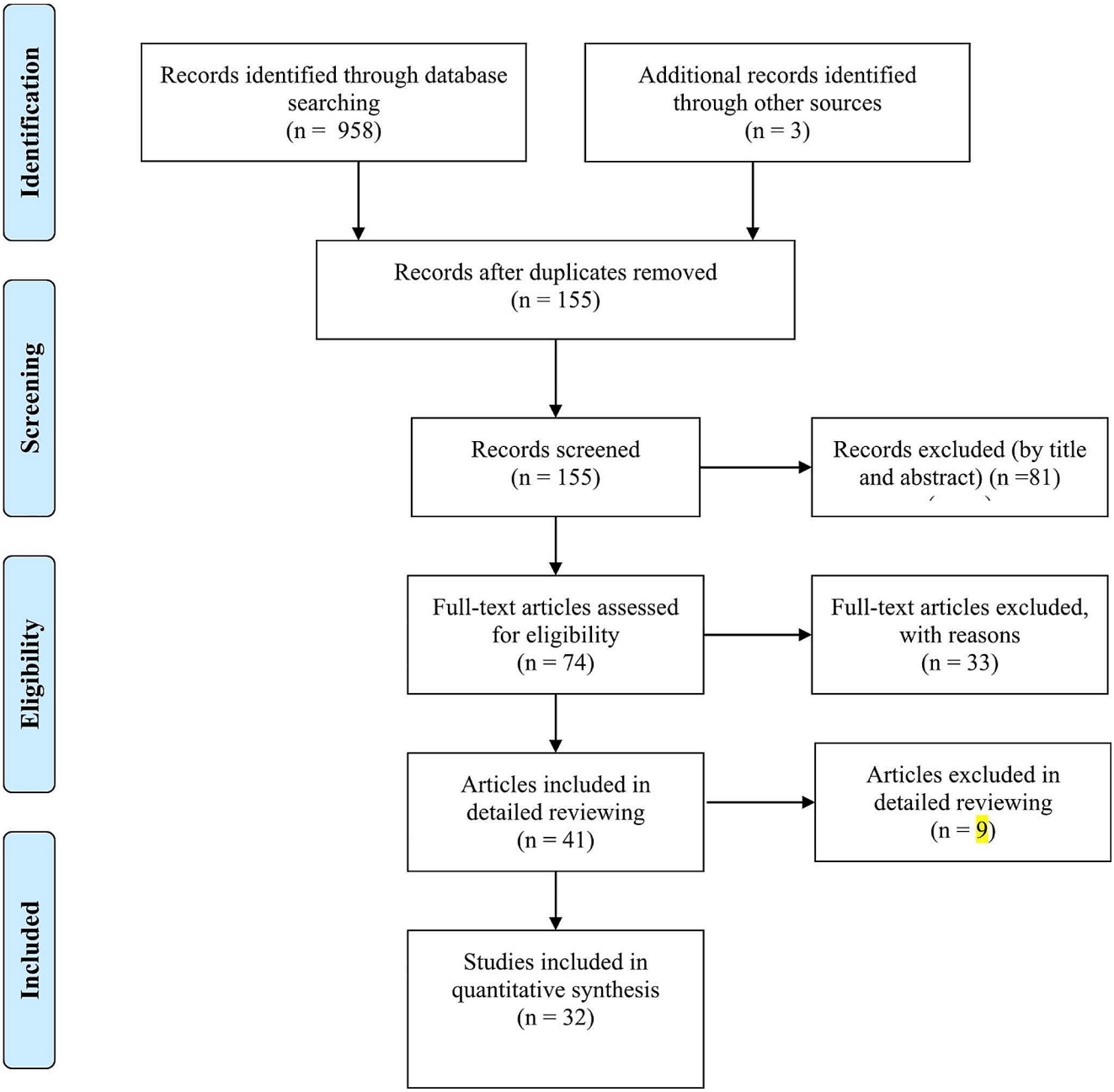


Fig. 1 PRISMA study search and selection flow diagram

The influence of pretreatment process on microbial community

Chemical pretreatment methods

Amongst the multifarious pretreatment approaches used to improve the anaerobic digestion process, the chemical pretreatments have drawn much attention for more solubilization of different substrates, especially, the recalcitrant ones [25]. The chemical pretreatment processes are commonly used to accelerate the hydrolysis process of lignocellulosic biomass, which are difficult to microbial degradation in anaerobic reactors [26]. The chemical pre-treatment including acid and alkaline condition, chemical addition and electrochemical process are most explored for anaerobically treatment of substrates rich with lignin, hemicellulose and cellulose content [27]. According to open literature, different chemical pretreatment process including Alkaline [28], thermo-alkaline [5], chemical addition [3], chemo-thermal [29], and electrochemical [15] were employed to enhance biogas potential and theirs effects on microbial community, dominant bacteria and archaea and mechanism in anaerobic digestion process. In this section, we focused on the effects of different chemical pretreatment process on corresponding dominant bacteria and archaea in anaerobic reactors.

Alkaline pretreatment Alkaline pretreatment process with pH higher than 10 are widely employed to enhance sludge hydrolysis and fermentation, sludge dewaterability and even pollutant removal [28]. As many physical, thermal and chemical pretreatments are questioned for intensive energy inputs and usage condition, alkaline pretreatment have drawn much attention due to having cost-efficient approach, easy application and few safety concern. Alkaline pretreatment can enhance the methane production via great production of short-chain fatty acids (SCFAs) in fermentation stages of anaerobic digestion process [30]. Calcium hypochlorite ($Ca(ClO)_2$) and NaOH as safety and cheap oxidants are widely employed for pretreatment process. These alkaline are able to destruct the cell components and improve the organic degradation and consequently enhance the methane production. For instance, Xuran Liu and et al. (2021) and Shu-Lin Wu et al. (2021) investigated the alkaline pretreatment with 2.0 M NaOH and ($Ca(ClO)_2$: 240 mg/g VS), respectively (See Table 1). The authors reported that these alkaline improve the BMP by 246.6 mL/g VS and 385.1 ± 3.3 mL/g VS, respectively [28, 31]. Although different other alkaline pretreatment methods have focused on improved BMP in anaerobic digestion such as Fe^{2+} and NaOCl [32], few studies surveyed the microbial population and microbial mechanism following the alkaline pretreatment in anaerobic digestion process [28, 31, 33]. Table 1 presents the predominant hydrolytic bacteria and methanogens identified in anaerobic reactors utilizing alkaline pretreatment. According to Table 1, *WSA2* were reported as the dominant archaea in the middle of anaerobic reactors pretreated with $Ca(ClO)_2$ [33]. Nevertheless, during the initial and final periods of the anaerobic reactors, *Methanosaeta*, which exhibited a low specific growth rate and a low half saturation coefficient, emerged as the dominant microorganism. Additionally, *WSA2* demonstrated the ability to produce methane through the reduction of methylated thiol, acetate, malonate, and propionate [34]. In addition, the predominance of *Bacteroidales* as the hydrolytic bacteria in anaerobic reactors proved that alkaline improved the quantity of SCFAs. *Bacteroidales* are able to degrade a wide variable of substrate to SCFAs [28]. In addition, Shu-Lin Wu et al. (2021) claimed that alkaline pretreatment with $Ca(ClO)_2$ led to reduced relative abundance *Chloroflexi* bacteria and *Euryarchaeota* archaea during the anaerobic digestion process [31]. The possible reasons for reduced relative abundance of bacteria and archaea can be considered as the specific topic of research. It can be attributed to high specific grow rate and need for high quantity of substrate in comparison with other microorganisms.

Thermo-alkaline pretreatment In recent years, the utilization of combined thermal and alkaline pretreatment has gained attraction as an effective method to disintegrate biomass and enhance methane production in the anaerobic digestion process [5]. The combined thermal and alkaline pretreatment can overcome the disadvantages of individual process such as high dosages of alkaline, inhibition the methanogens activity, needs for high amounts of acids to neutralize, inactivation the bacteria and high energy consumption [35, 36]. Alkaline pretreatment can facilitate the sludge disruption, solubility of organic matters, extracellular polymers extraction (EPS), conversion of the long-chain fatty acids to short-chain fatty acid (SCFAs) in great quantity and consequently improve the methane production [37, 38]. It is reported that alkaline pretreatment increased the solubility of sludge as order $NaOH > KOH > Mg(OH)_2$ and $Ca(OH)_2$ [4]. Hydrothermal pretreatment with high temperature of 60–180 °C are commonly combined to enhance the biomethane potential [4, 5, 35]. For instance, Jiaqi Fu et al. (2023) explored the thermo-alkaline pretreatment method to methane production in anaerobic digestion fed with co-substrate microalgae and sewage sludge. The authors reported that thermoalkaline pretreatment (%1.5 (W/V) NaOH and 90 °C for 120 min improved the BMP to 368.94 mL/g VS [5]. In addition, in another similar study, Van Toi Pham et al. (2020) reported that hydrothermal alkaline pretreatment (%10 W K_2C_3, 180 °C for 30 min) improved the BMP to 202 mL/gVS [4]. Although many studies have focused on improved

Table 1 Summary of predominant bacterial and archaeal composition in anaerobic digestion pretreated with different chemical methods

Pretreatment method	Pretreatment Time (min, h)	Reactor Type	Operation temperature (°C)	Inoculum	Substrate	BMP* (mL/gVS or ml/g COD)	Dominant Bacteria	Dominant Archaea	Ref.
Thermo-alkaline (%1.5 (W/V) NaOH, 90 °C)	120 min	Batch	37	Digested sludge	Sewage sludge + Microalgae	368.94 mL/g VS	*Firmicutes*, *Bacteroidetes*, *Euryarchaeota*, *Chloroflexi*	*Methanobacterium and Methanosarcina*	[5]
Thermo-Alkaline (1 N NaOH, 60 °C)	24 h	Batch	35	Digested sludge	Waste activated sludge	215.6 mL/g COD	*Bacteroidetes and Firmicutes*	*Methanosaeta and Methanobacterium*	[35]
Thermo-Alkaline (Urea: 2%, hydrothermal: 180 °C)	72 h	Batch	35 ± 1	Digested sludge	corn stover	205 mL/g VS	*Firmicutes*, *Bacteroidetes*	*Euryarchaeota*, *Halobacterota and Crenarchaeota*	[25]
Hydrohermal-Alkaline (%10 W K2C3, 180 °C)	30 min	Batch	35	Digested sludge	Sewage sludge	202 mL/gVS	*Bacteroidia*, *Proteobacteria*,	*Methanosaeta*, *Methanosarcina*	[4]
Alkaline (Ca(ClO)2, 0.01 g/gTSS)	30 min	Batch	35	Digested sludge	Thickened waste activated sludge (TWAS)	–	*Bacteroidales*, *Blvii28*	*WSA2 and Methanospirillaceae*, *Methanosaeta*, *and vadinCA11*	[33]
Alakaline (2.0 M NaOH, 2.0 M HCl)	12 days	Batch	35 ± 1	Digested sludge	PAM** flocculated sewage sludge (PFS)	246.6 mL/g VS	*Petrimonas and Sedimentibacter*	*Methanosaeta*	[28]
Alkaline (Ca(ClO)2: 240 mg/g VS)	48 h	Batch	37 ± 1	Digested sludge	waste activated sludge	385.1 ± 3.3 mL/g VS	*Chloroflexi*, *Bacteroidota*, *Proteobacteria*, *Firmicutes*, *Acidobacteriota*, *and Actinobacteriota*	*methanization (Euryarchaeota and Halobacterota)*	[31]
Chemical (Ca (OH)2: and % urea)	–	Batch	38	Digested sludge	corn stover	224.30 mL/g VS	*Acinetobacter*, *Jeotgalibaca*	*Comamonas and Methanobacterium*	[26]
Chemical (three-dimensional iron-carbon)	15 min	Batch	–	granular sludge	black liquor	181 mL/gCOD	*Proteobacteria and Bacteroidetes (Aminicenantales, Mesotoga)*	*Methanosaeta*	[41]
Chemical (CaO2, zero-valent iron (ZVI) (10 g/L)	24 h	Batch	37	Digested sludge	Waste activated sludge	–	*Anaerolineae*, *planctomycetia*, *bacteroidia*, *and Clostridia*	*Methanothrix*, *Methanospirillum*, *Methanolinea*, *and Methanobacterium*	[40]
Chemical (potassium ferrate (PF) and peroxymonosulfate (PMS))	–	Batch	35	Digested sludge	cow manure-cotton straw mixture	109.4 mL CH_4/g VS	*Sideroxydans*, *Herbinix*, *Clostridium*, *and Smithella*	*Methanosarcina and Methanobacterium*	[3]
Chemical addition (Glycerin)	–	Continuous	36	Digested sludge	Pig manure, Glycerin	507 mLCH_4/gVS	*Trichococcus focculiformis*, *T. palustris*	*Methanosaeta*, *Methanosarcina*	[42]
Chemical addition (Ferrate)	60 min	Batch	35	Digested sludge	WAS	–	*Macellibacteroides*, *Petrimonas*	–	[54]
Chemicals (H_2O_2)	–	Batch	37	Digested sludge	Wheat straw, Cattle manure (CM)	320 mLCH_4/gVS	*Gammaproteobacteria*, *Betaproteobacteria*	*Methanosarcina*, *Methanosaeta*	[55]
Chemo-Thermal (Fe 3+, 180 °C)	30 min	Batch	–	Digested sludge	Waste activated sludge	350.5 ± 9.6 mL-CH_4/g-COD	*DMER64*, *Proteiniphilum*, *Fermentimonas*	*Methanosaeta*, *Methanoculleus*, *and Methanoculleus*	[29]

Table 1 (continued)

Pretreatment method	Pretreatment Time (min, h)	Reactor Type	Operation temperature (°C)	Inoculum	Substrate	BMP* (mL/gVS or ml/g COD)	Dominant Bacteria	Dominant Archaea	Ref.
Chemo-Thermal (Iron powder, 170 °C)	–	–	37 ± 1	Digested sludge	Dewatered sludge	106.22 mL/gVS	–	*Methanomicrobia, Methanosarcinales*	[45]
Electrochemical (Electro-Fenton with iron electrode)	–	semi-continuous	37 ± 1	Digested sludge	waste activated sludge	135 ± 24 mL/g VS	*Caldilinea tarbellica and Caldilinea aerophila*	–	[15]

biogas production using the thermo-alkaline pretreatment processes, limited information are available on the effects of thermo-alkaline pretreatment on microbial community, dominant microbial population and microbial pathway [4, 5, 35]. Table 1 provides an overview of the predominant bacteria and archaea in the anaerobic digestion process facilitated by thermos-alkaline pretreatment. According to Table 1, the results showed that *Firmicutes*, *Bacteroidetes* are the most dominant hydrolysis bacteria in the anaerobic reactors pretreated with thermos-alkaline pretreatment process. Jiaqi Fu et al. (2023) surveyed the microbial community of anaerobic co-digestion process (sludge + microalgae) pretreated with thermoalkaline process using Illumina MiSeq platform. The authors reported that dominant phylum of bacteria are not affected by co-digestion and two supplementary substrates and *Firmicutes* due to ability to degrade the organic compounds were observed the most dominant bacteria in trial experiments [5]. In addition, in line with this study, Jungsu Park et al. (2022) claimed that thermo-alkaline pretreatment facilitated the growth of *Bacteroidetes* and *Firmicutes* as two dominant phyla of hydrolytic bacteria [35]. *Bacteroidetes* are able to degrade the protein in hydrolytic process and anaerobic digestion [35]. In addition, the dominant archaea observed in anaerobic reactors pretreated with thermo-alkaline process are summarized in Table 1. According to Table 1, *Methanosarcina*, *Methanobacterium*, and *Methanosaeta* are the dominant methanogenic bacteria in anaerobic reactors assisted with thermo-alkaline pretreatment process. As per the dominance of these methanogens, it can be concluded that the methane production in anaerobic reactors assisted with thermo-alkaline follows both acetoclastic and hydrogenotrophic pathway. Under anaerobic conditions, *Methanosarcina* is capable of utilizing a diverse range of substrates such as H_2/CO_2, methanol, acetate, and CO, and subsequently producing methane. While, *Methanosaeta* is a strict acetoclastic methanogen [39]. Furthermore, Van Toi Pham et al. (2021) surveyed the microbial dynamics of anaerobic digestion process assisted with thermo-alkaline process in different reaction time (30, 60, and 90 min) [4]. The taxonomic analysis indicated that *Methanosarcinales* are dominant archaea in different reaction time of thermo-alkaline process. Nonetheless, as the reaction time increased, *Methanosaeta* exhibited a decrease in population. Specifically, the relative abundance of this particular methanogen declined from 79.2 to 41.4% when the reaction time extended from 30 min to 90 min [4]. Moreover, the variation of alkaline used in thermo-alkaline pretreatment process and their effects on the microbial composition are suggested to further attention in the future.

Chemical addition Chemical addition is another pretreatment method used recently for enhanced methane pro-

duction from anaerobic reactors. In this content, different chemicals with low biotoxicity and capable of conversion the lignocellulosic biomass to ethanol and lipid, and activated carbon or iron powder to produce hydroxyl radicals (•OH) are commonly employed in hydrolysis stage of anaerobic digestion [3, 26, 40, 41]. Table 1 summarizes the effects of using different chemical addition on the BMP potential and corresponding microbial dominance in anaerobic digestion. According to Table 1, different chemicals including combined Ca $(OH)_2$ and Urea, three-dimensional iron-carbon, combined CaO_2, zero-valent iron, combined potassium ferrate (PF) and peroxymonosulfate (PMS), Glycerin and H_2O_2 have been used to both improved BMP and microbial community. For instance, Rodríguez-Abalde et al. (2019), reported that addition the Glycerin to anaerobic digestion fed with Pig manure improved the BMP to 507 mLCH_4/gVS [42]. In addition, as summarized in Table 1, the results indicated that *Methanobacterium* and *Methanosaeta* are the dominant archaea in different anaerobic reactors assisted with chemical addition as pretreatment process. As per the results, it can be concluded that methane production in anaerobic reactors assisted with chemical addition as pretreatment follows both hydrogenotrophic and acetotrophic methanogenic pathways. For instance, *Methanobacterium* as strictly hydrogenotrophic methanogens with capable of methane production via utilizing the both hydrogen and carbon dioxide was reported as the dominant methanogens in anaerobic reactors assisted with combined Ca $(OH)_2$ and urea, CaO_2, and zero-valent iron (ZVI) [26, 39, 40]. Furthermore, in anaerobic reactors utilizing a three-dimensional iron-carbon pretreatment process, *Methanosaeta*, known for its ability to metabolize acetate by reducing methyl carbon and oxidizing carboxyl into CO_2, was identified as the predominant acetotrophic methanogen [41]. However, according to Table 1, the dominant bacteria in this content are variable. For instance, Xiang Zheng et al. (2022) reported *Mesotoga* as the most dominant bacteria in anaerobic reactor fed with black liquor and assisted with three-dimensional iron-carbon [41]. *Mesotoga* is known as a typical bacterium in anaerobic reactors and capable of enhancing the fermentation and hydrolysis of recalcitrant substances like aromatic compounds [43]. While *Clostridium* with ability to crystalline cellulose degradation and TVFA production [44] were reported as dominant bacteria in anaerobic reactor assisted with CaO_2 and zero-valent iron (ZVI) pretreatment [40].

Thermo-chemical Although thermal pretreatment have been employed for enhanced methane production from anaerobic digestion, the main drawback imposed for this pretreatment process is production of Maillard reaction products (MRPs) [45]. MRPs are produced as a results of reaction between protein and reducing sugars during the thermal pretreatment. Native microorganisms face challenges in degrading MRPs during both anaerobic and aerobic processes, which can lead to adverse biological effects, including genotoxicity, cytotoxicity, and antimicrobial activity.

Generally, according to [46], Maillard reaction follows a three stages: (1) condensation of protein and reducing sugar (Shiff base), (2) dehydration and fragmentation of sugar moiety and Strecker degradation of amino acid moiety and generation the intermediated including ketones, aldehydes, furfural, etc., and (3) polymerization and condensation of intermediated to form MRPs. MRPs, also known as melanoidins, brown nitrogenous polymers and copolymers, or humin substances, are responsible for generating the brown coloration; color reaction occurred between protein and reactive carbohydrates which finally lower the biodegradability of Protein [47]. Over the recent years, many researchers attempted to lower the effects of intermediates having strong reducing power. Addition of the oxidizing reagents such as Ferric ion (Fe^{3+}45, 48]. For instance, Xiuqin Cao et al. (2022) reported that chemo-thermal pretreatment (Iron powder, 170 °C) led to production of 106.22 mL CH_4/gVS [45]. According to Table 1, only two studies have focused on the effects of thermo-chemical on microbial community of anaerobic digestion assisted with thermo-chemical pretreatment processes [29, 45]. For instance, Yi-Kun Geng et al. (2022) investigated the dominant bacteria at genus level in anaerobic digestion fed with waste activated sludge (WAS) and assisted with thermos-chemical (Fe^{3+}, 180 °C) pretreatment process [29]. The authors reported that *Proteiniphilum* and *Fermentimonas* with capability of protein degradation [49] experienced the highest relative abundance in anaerobic reactors. In addition, as per Table 1, it can be concluded that the methane production depending on substrates follows either hydrogenotrophic or acetoclastic mechanism pathway. Moreover, the higher relative abundance of *Methanoculleus* as dominant methanogens in anaerobic reactors pretreated with Chemo-Thermal (Iron powder, 170 °C) proves an important position for methane production. *Methanoculleus* belong to order Methanomicrobiales can use hydrogen and formate or carbon dioxide as substrate and produce the methane [45].

Electrochemical Electrochemical process has recently grasped attention for enhance disintegration, solubilization and bioavailability of organic compounds in anaerobic digestion. The electrochemical pretreatment processes are commonly based on lysis of microbial cells and flocs destruction [50]. Electro-Fenton (EF), as a type of electrochemical process, taking advantages of some reagents

(H_2O_2 and Fe^{2+}) in-situ is known as a clean technology for efficient recalcitrant organic degradation [51]. To the best of our understanding, there is only one study that specifically investigates the improved BMP (biomethane potential) and microbial community resulting from the implementation of EF (electro-Fenton) as a pretreatment process [15] **(See** Table 1**).** The authors reported that EF led to enhanced BMP by 135 ± 24 mL/g VS in anaerobic reactor fed with waste activated sludge (WAS). In addition, the authors claimed that the dominance of *Clostridium* are attributed to ability of this hydrolytic bacteria to convert the cellulose, proteins and lipids into TVFA. Moreover, *Clostridium* are able to produce the precursors of methane for methanogens in acetogenesis stage of anaerobic digestion [52, 53].

Biological pretreatment methods

Over the recent years, biomass-based renewable energy technologies such as AD have grasped much attention by researchers and authorities due to high efficiency, eco-friendly, practical, sustainable characteristics in order to produce the clean energy [56]. However, application the efficient and low-cost pretreatment is another issues imposed about the production the renewable energy from AD. Environmental-friendly and low-cost are the main driving force for researchers to use biological pretreatment process; the authorities attempted to find alternative for many pretreatment methods including physical and chemical due low-technical feasibility, high operation cost and toxic by-product and waste [16].

Liquid fraction of digestate and temperature-phased anaerobic digestion Liquid fraction of digestate (LFD) is known as a desirable techno-economical pretreatment method which are capable of parallel ammonolysis reaction and biotransformation of lignocellulosic material in order to enhance the BMP in AD. LFD are known with inorganic substances (ammonia, NO_3^- -N and potassium ions) and soluble organic matter (amino acids, proteins and sugars). LFD can aid the growth of high amount of microorganisms and even stabilize the AD via adjusting substrate C/N ratio [57, 58]. For instance, Yue Liu and et al. (2021) reported that LFD for 3 days improved the BMP generated from AD fed with corn straw [17]. In addition, temperature-phased anaerobic digestion (TPAD) due to excellent reactor stability, methanogen, abundance and diversity of the microbial community has recently employed for improvement the hydrolysis and BMP from anaerobic reactors [59]. For example, Mekdimu Mezmir Damtie and et al. (2021) investigated the TPAD in different ranges of temperature and the results indicated that operational temperature at 55 ˚C led to superior performance in terms of BMP, TVFA production and microbial activity compared to 85 ˚C [60].

Table 2 provides a summary of the impact of various biological pretreatment methods on microbial population and the prevalence of dominant bacteria and archaea in anaerobic digestion (AD) systems utilizing different substrates. As indicated in Table 2, *Bacteroidetes* and *Firmicutes* are the primary bacterial phyla observed in AD processes aided by biological LFD and TPAD pretreatment methods [1, 17, 60]. These syntrophic bacteria are able to secrete hydrolytic extracellular enzymes and convert the organic matter into TVFA, in particular, acetic acid [61, 62]. In case of archaeal communities, the phylum of *Euryarchaeota* and *Methanobacteriaceae*, *Methanosaetaceae* and *Methanosarcinaceae* genera were reported the dominant in AD assisted with LFD pretreatment process [17]. These results indicated that LFD improve the methane production and follows both acetolactic and hydrogenotrophic methanogenesis [63].

Enzymatic pretreatment methods Enzymatic pretreatment is known as a promising and viable technology to improve the BMP in anaerobic digestion process [10]. Enzymatic pretreatment due to some interesting advantages including easy operation, eco-friendly characteristics and mild reaction condition can convert EPS as the bottleneck in AD into monosaccharides, fatty acids, or amino acids. Enzymatic pretreatment improves the breaking of glucosidic, ester or peptide bonds in macromolecule compound and even enhance the destroying the microbial cells by releasing intracellular substances into environment [64, 65]. To best of our knowledge and as per search strategy, just one study has focused on both the improvement of BMP in AD assisted with cocktails of enzymes (400 U/g cocktail enzymes) and their corresponding microbial population and dominant bacteria and archaea [10]. Table 2 summarizes the dominant bacteria and archaea observed in AD assisted with enzymatic pretreatment and fed with sewage sludge. As seen in Table 2, *Fermentimonas* (belong to *Bacteroidetes* Phylum) were observed with highest abundance in AD assisted with E strategy (enzymes and enhancer) pretreatment process. *Fermentimonas* are able to produce acetic acid, H_2 and CO_2 via degradation of carbohydrates and complex proteinaceous compounds [66]. In addition, the genus *Methanosaeta* was reported as the most abundant methanogenisis in AD assisted with E strategy [10].

Microaerobic pretreatment method Over the recent years, there are controversy on application of microaerobic (MA) (DO = 0.1-1 mg/L) in anaerobic digestion process [67]. On the one hand, MA are able to improve the degradation of lignocellulosic substances via functional activities of both facultative and aerobic microorganisms, leading to enhance solubilization. On the other hand, many researchers claimed

Table 2 Summary of predominant bacterial and archaeal composition in anaerobic digestion pretreated with different biological methods

Pretreatment method	Pretreatment Time (min, h)	Reactor Type	Operation temperature (°C)	Inoculum	Substrate	BMP* (mL/gVS or ml/g COD)	Dominant Bacteria	Dominant Archaea	Ref.
Liquid fraction	3 days	Batch	35 ± 2 °C	Digested sludge	swine manure (SM), food waste (FW), and wheat straw (WS)	277.9 mL/gVS	*Bacteroidetes*, *Firmicutes*	–	[1]
Liquid fraction	3 days	Batch	35 ± 1 °C	Digested sludge	corn straw	265.1 mL·g- VS^{-1}	*Firmicutes and Bacteroidetes*	*Methanobacteriaceae*, *Methanosarcinaceae and Ruminococcaceae*	[17]
temperature-phased anaerobic digestion (TPAD)	908 h	semi-continuous	35	Digested sludge	*Chlorella vulgaris* microalgae	–	*Firmicutes and Proteobacteria*	–	[60]
Enzyme and Trace elements (400 U/g cocktail enzymes and 1.24% trance elements enhancers)	6 h	Batch	35	Pig manure	Sewage sludge	83.82 mL g^{-1} TS^{-1}	*Fermentimonas and Lutispora*	*Methanosaeta*	[10]
Microaeration (DO: 0.1-1 mg/L)	12	Batch	35	Fresh cow manure	Dry corn stover	166.1 mL/g TS	*Firmicutes*, *Bacteroidota*, and *Proteobacteria*	–	[20]

that addition of oxygen into AD cause deteriorate effects on functional activity of methanogenis and consequently reduced methane production [20, 68]. Therefore, two factors should be considered for using MA as a pretreatment process to maximize the BMP in anaerobic reactor: pretreatment time should not be in a long-time and oxygen respiration is needed to be regulated in low concentration. In addition, the pH of solution is also needed to regulate during the MA pretreatment [69]. It is important to note, there are two strategies to use MA: MA as a pretreatment in a separate reactor and regulation the AD with DO = 0.1-1 mg/L during the incubation period [20]. Although, many studies have focused on MA in AD [70–73], only one study has focused on the improvement of BMP in AD assisted with MA and corresponding microbial population and dominant bacteria and archaea [20]. According to Table 2, Haipeng Xu and et al. (2021) reported that *Firmicutes*, *Bacteroidota*, and *Proteobacteria* were the dominant bacteria in AD assisted with MA [20]. The high relative abundance of these facultative bacteria are attributed to MA, providing conditions for growth of these hydrolytic bacteria in AD. *Firmicutes*, *Bacteroidota*, and *Proteobacteria* play important role in hydrolysis of macromolecular and organic acid production [74].

Physical pretreatment methods

Thermal pretreatment method Over the recent year, thermal hydrolysis pretreatment (THP) with different ranges of temperature and pressure have been employed for hydrolysis the lignocellulosic polymers and other complex compounds in different substrates [18, 75]. THP is recognized as a highly effective and promising technique for breaking down recalcitrant compounds, enhancing sludge stabilization, and improving methane production in the anaerobic digestion process [76]. Recently, THP in different forms of pretreatment including hydrothermal pretreatment (known as eco-friendly technique due to chemical-free nature) and with chemical compound and combined with ultrasonic have been used to enhance the methane production in anaerobic digestion [4, 77, 78].

For instance, Yufang Wei and et al. (2022) reported that hydrothermal process with temperature of 140 °C for 20 min improved the BMP by 590.6 ± 8.0 mL.gVS^{-1} in AD fed with food waste (FW) and Enteromorpha [78]. In addition, Huihui Chen and et al. (2021) investigated the BMP in AD fed with sewage sludge using the Sonothermal (Ultrasonic: 26 kHz; 2000 kJ $kgTS^{-1}$, thermal: 55 °C) within 8 h and reported that the highest BMP was 0.27–0.31 m^3 CH_4 /$kgCODrem^{-1}$ [77]. According to open literature review, some studies have focused on different types of THP on the microbial population and microbial dynamics in AD [18, 75, 77–80]. Table 3 summarizes

Table 3 Summary of predominant bacterial and archaeal composition in anaerobic digestion pretreated with different thermal methods

Pretreatment method	Pretreatment Time (min, h)	Reactor Type	Operation temperature (°C)	Inoculum	Substrate	BMP* (mL/gVS or ml/g COD)	Dominant Bacteria	Dominant Archaea	Ref.
Thermal (170 °C, 10 bar)	30 min	Batch	37 ± 1	Digested sludge	swine manure	96.0 ± 37.1 mL-CH_4/g-VS/day	*Firmicutes*, *Bacteroidetes*	*Bathyarchaeota*, *Euryarchaeota*	[18]
Thermal (70 °C)	3 days	Batch	35	Digested sludge	Raw swine manure	160 mLCH_4/gVS	*Bacteroidales* *Clostridiales*	Methanomassiliicoccaceae, Methanobacteriaceae	[83]
Hyrothermal (140 °C)	20 min	two-stage AD	–	Anaerobic Digestate	food waste (FW) and Enteromorpha	590.6 ± 8.0 mL·gVS^{-1}	*Firmicutes*	*Methanosarcina*, *Methanobacterium*, *Methanosaeta*, and *Methanoculleus*	[11]
Hyrothermal (160 °C)	60 min	Batch	37	Digested sludge	erythromycin fermentation dregs	428.3 ml g^{-1} VS	*Aminicenantales* and *Sedimentibacter*	*Candidatus_Methanofastidiosum* and *Methanosarcina*	[75]
Hydrothermal (170–175 °C)	60 min	Two-phase	37	Digested sludge	WAS, OFMSW	0.71 L biogas/g VSadded	*Firmicutes*,*Bacteroidetes*	–	[84]
Sonothermal (Ultrasonic: 26 kHz; 2000 kJ kgTS-1, thermal: 55 °C)	8 h	Continuous	37	Digested Sludge	sewage sludge (Primary sludge: secondary sludge = 40:60% (volume))	0.27–0.31 m^3 CH_4 / $kgCODrem^{-1}$	*Firmicutes* (23.5%), *Thermotogae* (23.5%)	*Methanosarcinales* (*Methanosaeta*)	[77]
ChemoThermal (CPAM(coagulant): 20 mg/gTS, thermal: 170 °C)	30 min	Batch	37	Digested Sludge	sludge	38.9 ml CH_4/g TS	*Firmicutes* (50.0-61.2%), *Bacteroidetes* (11.4–17.9%)	*Methanosarcina* sp. *and* *Methanobacterium* sp	[79]

BMP: Biomethane Potential ** PAM: polyacrylamide

Table 4 Summary of predominant bacterial and archaeal composition in anaerobic digestion pretreated with ultrasonic pretreatment methods

Pretreatment method	Pre-treat-ment Time (min, h)	Reactor Type	Oper-ation tem-pera-ture (°C)	Inoculum	Substrate	BMP* (mL/gVS or ml/g COD)	Dominant Bacteria	Dominant Archaea	Ref.
Ultrasound (1 W/mL, 25 kHz) + Chem-ical (potassium ferrate)	15 min	Batch	35	Digested Sludge	Raw sludge	228.83 ± 4.76 mL/g VSS	*Bacteroidetes*	*Methanosaeta and Candidatus_Methanofas*	[19]
Ultrasonication	–	–	–	–	FW, Livestock waste	–	*Proteobacteria*, *Bacteroidetes*	*Methanosaeta*, *Methanotorris*	[93]
Ultrasonication	16 min	Semi-contin-uous	–	Digested sludge	WAS	0.27 L/g VS day	*Bacteroidetes*, *Proteobacteria*	*Methanosaeta*, *Methanobrevibacter*	[94]

the results obtained from the different studies focused on dominant bacteria and archaea in AD assisted with a variety types of THP and fed with different substrates. According to Table 3, *Firmicutes* play important role in hydrolysis of substrates in AD assisted with THP. It is believed that although THP technique with high temperatures kill the various microorganisms, *Firmicutes* are resistant to different environmental disturbance due to ability to produce the endospores [81]. As previously stated, *Firmicutes* play a role in facilitating the improved hydrolysis of cellulosic biomass for acetogenesis [82].

Ultrasonic pretreatment method As earlier mentioned, the pretreatment processes of anaerobic digestion aims to enhanced solubilization, methane production, sludge stabilization, resource recovery and treatment [2]. However, extracellular polymer substances (EPS), recalcitrant organic matters and microbial cell membrane are the most known obstacles which inhibit the microbial activity in anaerobic digestion [85]. Ultrasound (US) as a physical pretreatment due to cavitation mechanism can improve the biodegradability of organic matters in biological reactors. In US pretreatment, the temperature increase to 5000 K and the microbubble with ability of production highly reactive OH radicals were generated following some consequent stages: growth bubble until unstable diameter and the collapse. In addition, the other beneficial effects of US are enzyme extraction, pollutant removal and intracellular material recovery [86]. Generally, according to frequency, US are divided into three regions: power ultrasound (20–100 kHz), high frequency ultrasound (100 kHz– 1 MHz) and diagnostic ultrasound (1–500 MHz) [87]. However, according to open literature, the most efficient sludge disintegration are achieved in US with frequency of 20–40 kHz [88]. For instance, Bing Guo et al. (2021) reported that the combined pretreatment US (1 W/mL, 25 kHz, 15 min) and potassium ferrate (PF) (0.05 g/g TSS) improved the BMP by 228.83 ± 4.76 mL/g VSS in anaerobic reactor fed with WAS [19]. Table 4 summarizes the dominant bacteria and archaea in anaerobic reactors assisted with US fed with different substrates. According to Table 4, all three studies focused on dominant bacteria in anaerobic reactors assisted with US reported that *Bacteroidetes* and *Proteobacteria* were the most dominant hydrolytic bacteria [19, 89, 90]. The enrichment of *Bacteroidetes* in anaerobic reactors are attributed to ability of these syntrophic bacteria to hydrolysis of the protein and transformation the organic matters to VFA. *Proteobacteria* are known as the main acetate consumer in process of methane production via acceotolastic methane production pathway [90, 91]. In addition, the *Methanosaeta* were reported as the methanogens with higher abundance in anaerobic reactors assisted with US pretreatment (See Table 4). According to Results, it can be concluded that the increased abundance of syntrophic bacteria and *Methanosaeta* spp., as the acetate-utilizing methanogens are the main reason for enhanced BMP in anaerobic reactors assisted with US pretreatment [92].

Conclusion

In this study, our objective was to investigate the impact of various pretreatment methods on the BMP and microbial community in AD systems using different substrates. In order to address the question effectively, an extensive search strategy was implemented across various electronic databases, marking the first instance of such an approach to identify studies specifically addressing the aforementioned question. The results indicated that advent of new and modern molecular technologies improve the knowledge on microbial community in AD. In addition, the results obtained from the present research indicated that different chemical, biological, and physical pretreatment technologies improve the BMP. In addition, the pretreatment method play important role in dominance of hydrolysis bacteria and archaea tolerant to operational and environmental conditions. Moreover, hydrogenotrophic bacteria due to high resistance to environmental conditions were the

main pathways of biogas production AD assisted with different pretreatment methods. In general, an in-depth understanding of the composition and mechanisms of the dominant microorganisms within the bacterial and archaeal communities can provide valuable insights for improved microbial management in AD. This knowledge encompasses aspects such as enhancing stability and optimizing the performance of AD processes.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Consent to participate The authors agree to participate in this research study.

Consent for publication All authors agreed on the publication.

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